

Machine Learning

Exercise 1

EXERCISE 1

Consider the linear function:

$$y_1(x_1, x_2) = 2 - \frac{2}{3}x_1 - x_2$$

- Draw in the (x_1, x_2) -coordinate system the hyperplane defined by $y_1(x_1, x_2) = 0$.
- Draw the vector $\mathbf{w} = (-\frac{2}{3}, -1)$ and verify that this vector is orthogonal to the hyperplane
- What is the decision region defined by $y_1(x_1, x_2) > 0$?

Solution:

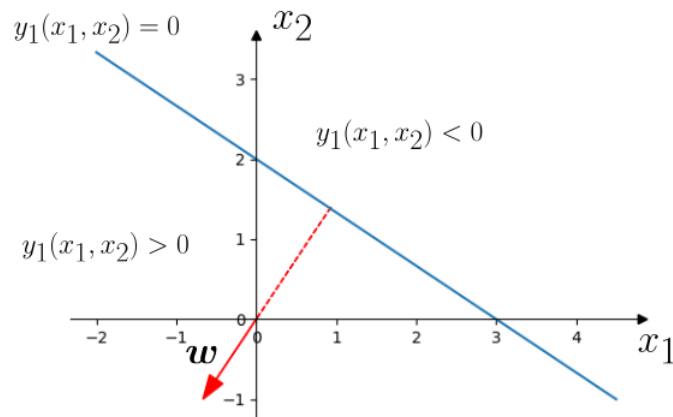


Figure drawn using Matplotlib. The actual vector $\mathbf{w}$ is the solid arrow. The dashed line extends it in the opposite direction to demonstrate the right angle with the hyperplane $y_1(x_1, x_2) = 0$ (blue line). Note that when using plotting software the angles will only come out right when both axes are equally scaled, i.e., the distance between two 'ticks' must be the same on the x_1 and the x_2 axis.

EXERCISE 2

We consider the example from lecture slide 1.32 (I use X.YY to denote the YY'th slide in lecture no. X).

- Compute the sum-of-squares errors of the two models $\mathbf{W}_1$ and $\mathbf{W}_2$.
- Find a solution $\mathbf{W}_3$ that has both 100% accuracy and a low sum-of-squares error.

Solution:

Sum-of-squares error for $\mathbf{W}_1$: 178, for $\mathbf{W}_2$: 47. One way to preserve the accuracy of $\mathbf{W}_1$ while reducing the SSE is to reduce the value of $w_{1,1}$ from 10 to a smaller value, because the 10 obviously was responsible for a large part of the SSE we get for $\mathbf{W}_1$. If we set the first column of $\mathbf{W}_3$ to $(1, 0)^T$, then $\mathbf{W}_3^T \mathbf{X}$ will already get the first row of $\mathbf{T}$ perfectly right. If we now set the second column to $(0, 0.1)^T$, then we get

$$\mathbf{W}_3^T \mathbf{X} = \begin{pmatrix} 1 & 1 & 0 \\ 0.1 & 0.5 & 0.1 \end{pmatrix}$$

which still gives us 100% accuracy and an SSE of 1.07. This is a somewhat heuristic construction of $\mathbf{W}_3$, and there can be many other solutions. We can also solve the matrix equation for the optimal solution (but this is no longer an easy pen and paper exercise), and we get

$$\mathbf{W}_{opt} = \begin{pmatrix} 1 & -0.333 \\ 0 & 0.111 \end{pmatrix}$$

which gives

$$\mathbf{W}_{opt}^T \mathbf{X} = \begin{pmatrix} 1 & 1 & 0 \\ -0.222 & 0.222 & 0.111 \end{pmatrix}$$

which is 100% accurate, and achieves an SSE of 0.888.

EXERCISE 3

In addition to y_1 from Exercise 1, consider

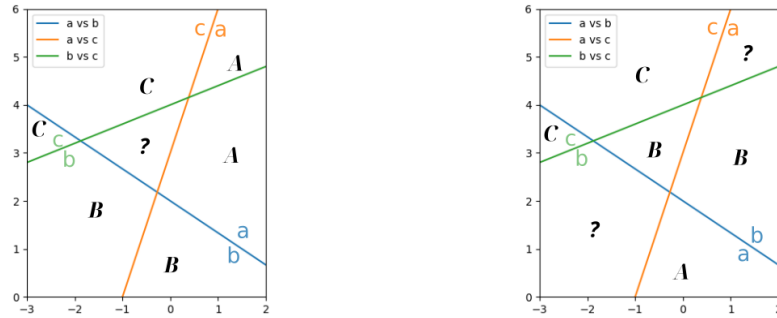
$$\begin{aligned} y_2(x_1, x_2) &= 3 + 3x_1 - x_2 \\ y_3(x_1, x_2) &= 1 + \frac{1}{10}x_1 - \frac{1}{4}x_2 \end{aligned}$$

- Assume that y_1, y_2, y_3 are three linear classifiers for the binary classifications “class a vs class b”, “class a vs. class c”, and “class b vs. class c” respectively. What are the decision regions for the resulting “one against one” classifier? Before you can answer this question, you will have to decide which decision regions are associated with which class, e.g., whether $y_1(x_1, x_2) < 0$ should mean that you classify (x_1, x_2) as a or b .

- Now assume that y_1, y_2, y_3 are three discriminant functions for the classes a, b, c . What are the decision regions for the resulting discriminant function classifier?

Solution:

This figure shows two possible solutions:



The colored, lower-case letters on the decision boundaries show the decision regions for the one-against-one classifiers. The left and right figure are different in that we have swapped for the a-vs-b classifier which side of the boundary is associated with class a , and which is class b . The black upper-case letters show the decision regions for the resulting multiclass classifier. Note that the undecided region is quite different in these two cases.

For the second part, we first determine the lines of (x_1, x_2) points where two of the three functions y_1, y_2, y_3 are equal. Equating the functions and solving for x_2 gives us:

